DAM - Introduction

LEIM – DAM

MOBILE APPLICATION DEVELOPMENT

DESENVOLVIMENTO DE APLICAÇÕES MÓVEIS

DAM

- DAM
  - Mobile Application Development / Desenvolvimento de Aplicações Móveis
  - 6th Semester of LEIM course
- Hours/week:
  - 4,5 hours = 1,5 theoretical-practical + 3 laboratory
- Credits:
  - 6 ECTS
- Requirements:
  - Modeling and Programming/Object Oriented Programming

Contacts

- Ana Correia – LEIM61N
  - Email: ana.correia@isel.pt
  - Office: F.0.40
- António Teófilo – LEIM62D – course unit coordinator
  - Email: antonio.teofilo@isel.pt
  - Office: F.0.14
- Pedro Fazenda – LEIM61D
  - Email: pedro.fazenda@isel.pt
  - Office: E.2.16.4

Learning outcomes

- **Mobile programming**, in Android and Kotlin
  - Develop Android applications using Kotlin and Jetpack Compose
  - Apply modern mobile software architectures
  - Integrate apps with local and remote data sources
  - Handle asynchronous operations
  - Produce, test and deploy mobile software
  - Use AI tools effectively in the produced software and in the development process

Assessment

- Assessment: distributed assessment without final exam
  - **35%** 4 Lab works (individual) and mandatory (minimum of: 8v in each lab; 9.5v in avg)
    - Submit, in moodle, a zip file with code
  - **65%** Final project + Report + Discussion (oral evaluation) (individual)
  - Minimal classification: 9.5v in avg of Lab Works and in Final Project/evaluation
    - Submit, in moodle, a zip file with a PDF report and code
- Lab Work evaluation:
  - **70%** Execution of the tutorial guide (use of Kotlin programming language is mandatory)
  - **30%** Extra features, added proactively and evaluated by difficulty, originality and quality
  - For each Lab work students must deliver their code and do an oral discussion
    - Code must be delivered up to the deadline date
    - After the deadline, a penalty of 0.75v (0 to 20v) per day is applied
    - Each lab should be discussed, in classroom, until the delivery date of the next lab
- No final special assessment (época especial), due to the nature of this course (distributed assessment and without a final exam) - RPAC

Labs and Final Project

- Labs topics
  - 1 - Android basics
  - 2 - Android events, layouts, internalization, fragments - Cool Weather App
  - 3 - JetPack compose, navigation, MVVM, KMP
  - 4 - Room, Ktor, KOIN

- Final Project
  - Design and Implementation of a free theme Android Application
  - Grade ranges:
    - Basic version of an existing app: up to 14v
    - Complete full version of an existing app, or an app inspired by an existing app: up to 18v
    - Original app: up to 20v
  - Evaluation will be based on code quality and complexity
    - A good and nice User Interface is a base requirement
  - Part A – Teacher approval is required to proceed to Part B
  - Part B – It must be a continuation of Part A

Schedule 2024/2025 Summer Semester

No class due to holydays:
- LEIM61D: 4h30
- LEIM62D: 1h30
- LEIM61N: 9h00

Delivers: at the end of the week

Week	Monday	Holidays/Other	Week topics	Delivers
1	16/02/2026	16/mon, 17/tue (61/2D)	Presentation	
2	23/02/2026		Lab 1: Android Basics (Hello World)	
3	02/03/2026			L1
4	09/03/2026		Lab 2: Android events, layouts, internalization, fragments - Cool Weather App	
5	16/03/2026			L2
6	23/03/2026		Lab 3: JetPack compose, navigation, MVVM, KMP	
7	30/03/2026	2/thu, 3/fri (**61N**) Easter	Final Project part A (FPA): project proposal	L3 & FPA
	06/04/2026	Easter	Easter holydays	
8	13/04/2026		Lab 4: Room, Ktor, KOIN	
9	20/04/2026	25/sat		L4
10	27/04/2026	1/fri (**61N**)	Final Project part B (FPBP): planning	
11	04/05/2026			FPBP
12	11/05/2026		Final Project part B(FPBD): development	
13	18/05/2026			
14	25/05/2026			
15	01/06/2026	4/thu		FPBD

Final oral discussions: 22/06 to 26/06

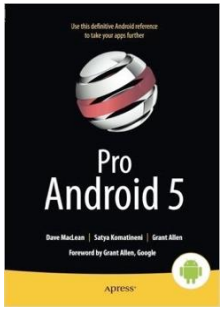
Kotlin and Android: Learn and work

- Kotlin
  - Learn:
    - Kotlin Documentation (Kotlin Docs)
      - https://kotlinlang.org/docs/home.html
  - Work: IDE: IntelliJ Idea - https://www.jetbrains.com/idea/

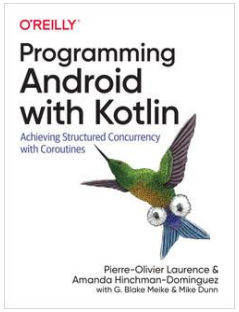
- Android
  - Learn:
    - Google Android Developers Website
      - https://developer.android.com
    - Other sources of information:
      - https://www.youtube.com/@PhilippLackner
  - Work: IDE: Android Studio - https://developer.android.com/studio

Bibliography - Android

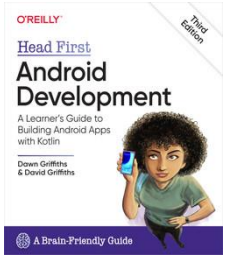
- **Pro Android 5**
  - Dave MacLean , Satya Komatineni , Grant Allen Apress
  - ISBN: 9781430246800



- **Programming Android with Kotlin**
  - Pierre-Olivier Laurence, Amanda Hinchman-Dominguez, Mike Dunn, G. Blake Meike
  - ISBN: 9781492063001



- **Android Development**
  - Dawn Griffiths, David Griffiths
  - ISBN: 9781492076476



Get prepared

- **Kotlin – install IntelliJ**
  - Build an Hello World in Kotlin, in console mode

- **Android Development – install Android Studio**
  - Build an Hello World Android APP in Kotlin